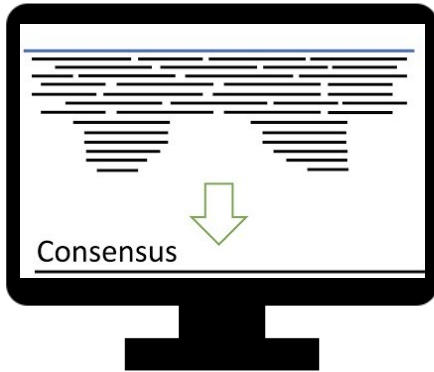


Disclaimer

- This training resource was developed to be used with the CZ ID platform
- In October, 2026, stewardship of CZ ID transitioned to UCSF and the platform was renamed to SeqtoID
- These training materials are not being actively maintained and may not reflect differences between CZ ID and SeqtoID, subsequent changes to the platform, or current methods and best practices in metagenomic sequencing and analysis

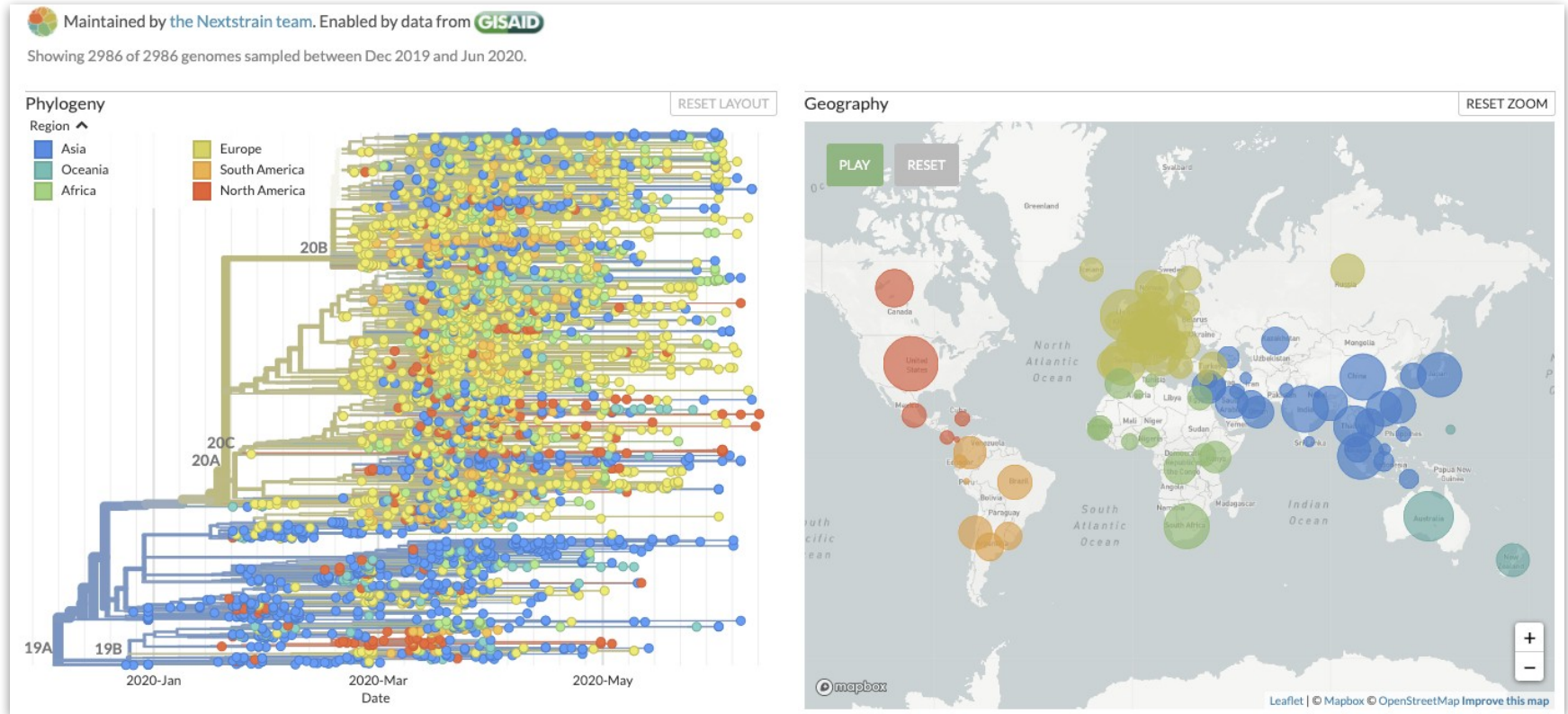
Virus Consensus Genomes



Outline

- Consensus genomes
- CZ ID's workflow for generating viral consensus genomes
- CZ ID's assembly report

Genome sequences are important! What we sequence are consensus genomes.



Consensus genomes: Represent the most frequently observed nucleotide at each site in the genome at the time of sample collection

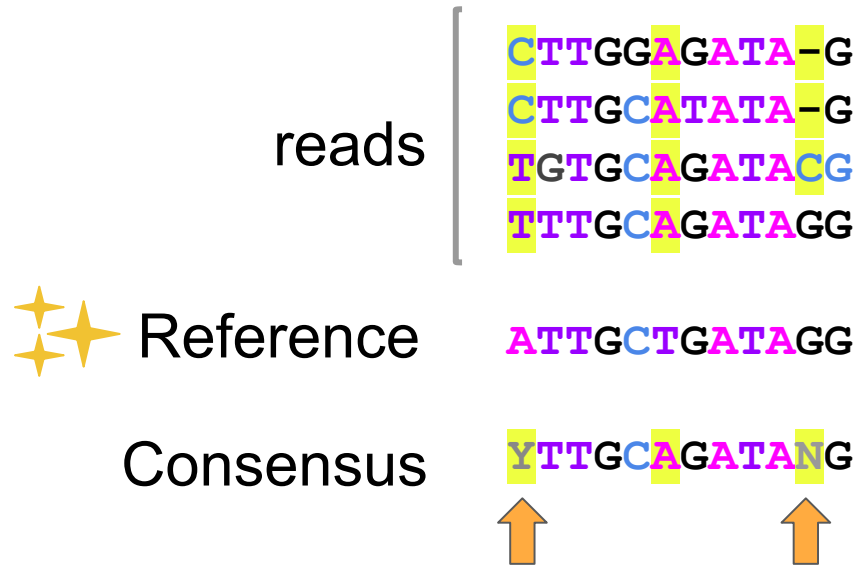


Consensus genomes: Some sites may display high diversity and the majority rule for calling a base cannot be applied. These sites can be defined with the ambiguity code.

IUPAC Nucleotide Code	Base
A	Adenine
C	Cytosine
G	Guanine
T (or U)	Thymine (or Uracil)

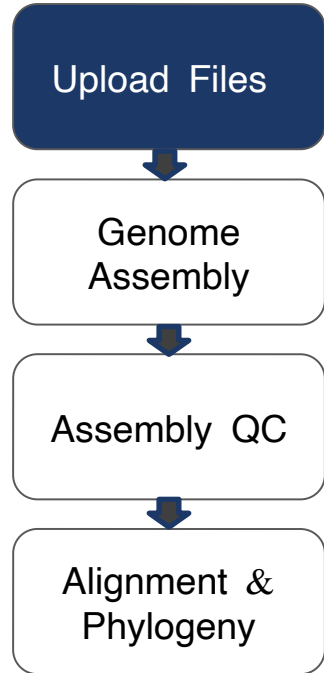
IUPAC Nucleotide Code	Degenerate Base
R	A or G
Y	C or T
S	G or C
W	A or T
K	G or T
M	A or C
B	C or G or T
D	A or G or T
H	A or C or T
V	A or C or G
N	Any base

Consensus genomes: Some sites may display high diversity and the majority rule for calling a base cannot be applied. These sites can be defined with the ambiguity code.



Ns - considered “missing data”

Workflow for Generating Consensus Genomes



Input Files

- **Sequencing file(s)** - Raw sequence reads in FASTQ format
- **Reference sequence** - Reference sequence you would like use to assemble the viral consensus genome by mapping reads against it (FASTA format).
 - You should select a reference sequence that is most similar to the genome sequence you are trying to assemble
- **Primer BED file** - File specifying primer positions when using viral sequence data obtained through primer spiking for target enrichment or PCR-based assays. Not required for data obtained through metagenomic sequencing.

Select Project

Project

Meharry MD Workshop Private Project

[+ CREATE PROJECT](#)

Analysis Type

- ☐  **Metagenomics**
Run your samples through our metagenomics pipeline. Our pipeline supports Illumina and Nanopore technologies.
- ☐  **Antimicrobial Resistance**
Run your samples through our antimicrobial resistance pipeline. Our pipeline supports metagenomics or whole genome data. It only supports Illumina. You can also run the AMR pipeline from within an existing project by selecting previously uploaded mNGS samples. You can check out the AMR pipeline on Github [here](#).
- ☒  **Viral Consensus Genome**
Run your samples through our Illumina supported pipeline to get viral consensus genomes using your own reference sequence. Pipeline report does not link to Nextclade.

Pipeline Version:
3.5.0
The selected project uses the above version to run your samples. [Learn More](#)

Taxon Name ⓘ
Select Taxon Name ▼

Reference Sequence ⓘ
[SELECT FILE](#)

Trim Primers — Optional ⓘ
[SELECT FILE](#)
- ☐  **SARS-CoV-2 Consensus Genome**
Run your samples through our Illumina or Nanopore supported pipelines to get consensus genomes for SARS-CoV-2. Send consensus genomes to Nextclade.

Select Files

Your Computer Basespace

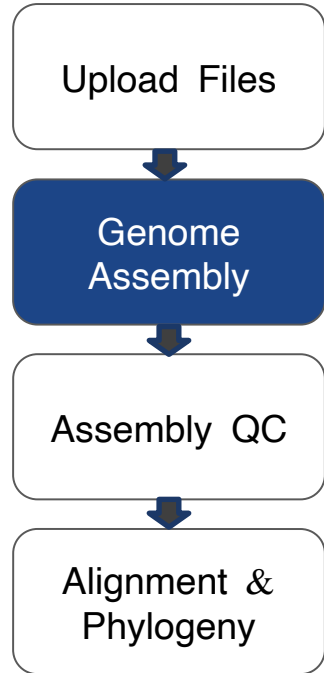
Select Input Files [MORE INFO](#)

Drag and drop your files here, or [click to use a file browser](#).

Continue

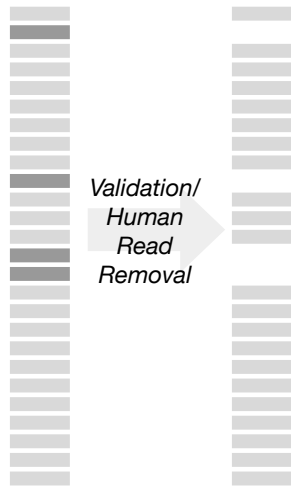
Cancel

Workflow for Generating Consensus Genomes



Genome Assembly Pipeline

Raw reads

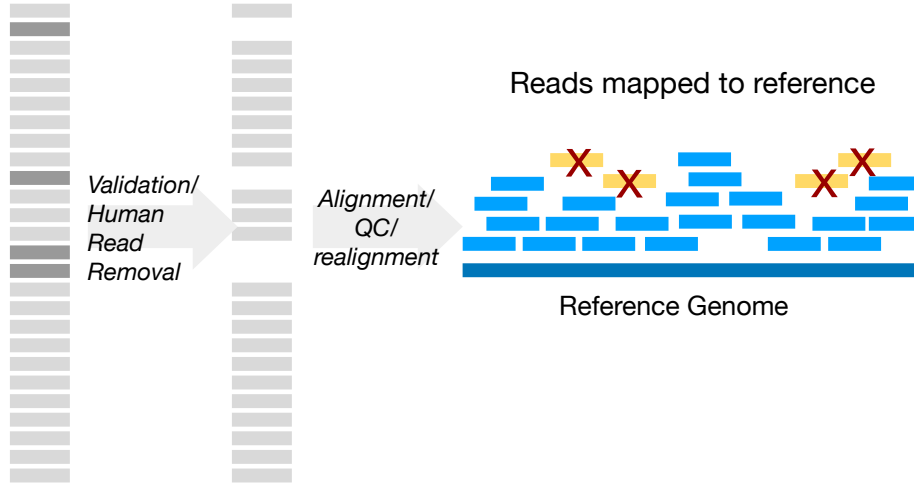


Notes:

- Required files
 - Raw reads (Illumina FASTQ files)
 - Reference sequence (FASTA format)
 - BED primer file (Optional)
- Reads capped at 50M
- Pipeline only removes human host reads

Genome Assembly Pipeline

Raw reads

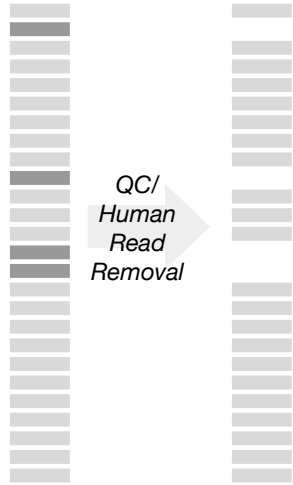


Notes:

- Alignment software: [minimap2](#)
- Mapped reads undergo QC; software: [Trim Galore](#)
 - Trim low quality bases (<Q20)
 - Trim adapters
 - Trim short reads (< 20bp)
- After QC, remaining reads are remapped to reference (minimap2)

Genome Assembly Pipeline

Raw reads

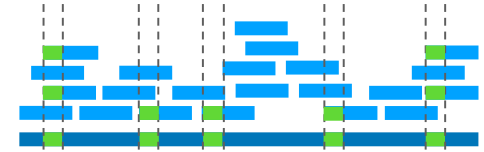


Alignment

Reads mapped to reference



Trim Primers

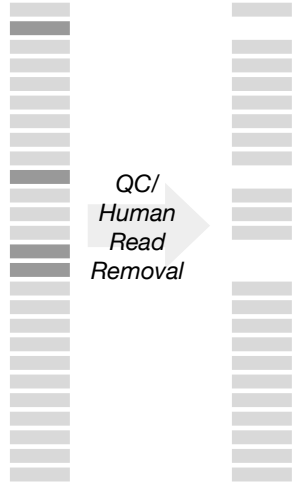


Notes:

- Software for trimming primers (soft-clipping): [iVar](#)

Genome Assembly Pipeline

Raw reads



QC/
Human
Read
Removal

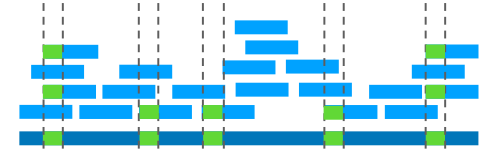
Alignment

Reads mapped to reference



Reference Genome

Trim
Primers



Consensus Sequence



Call
Consensus

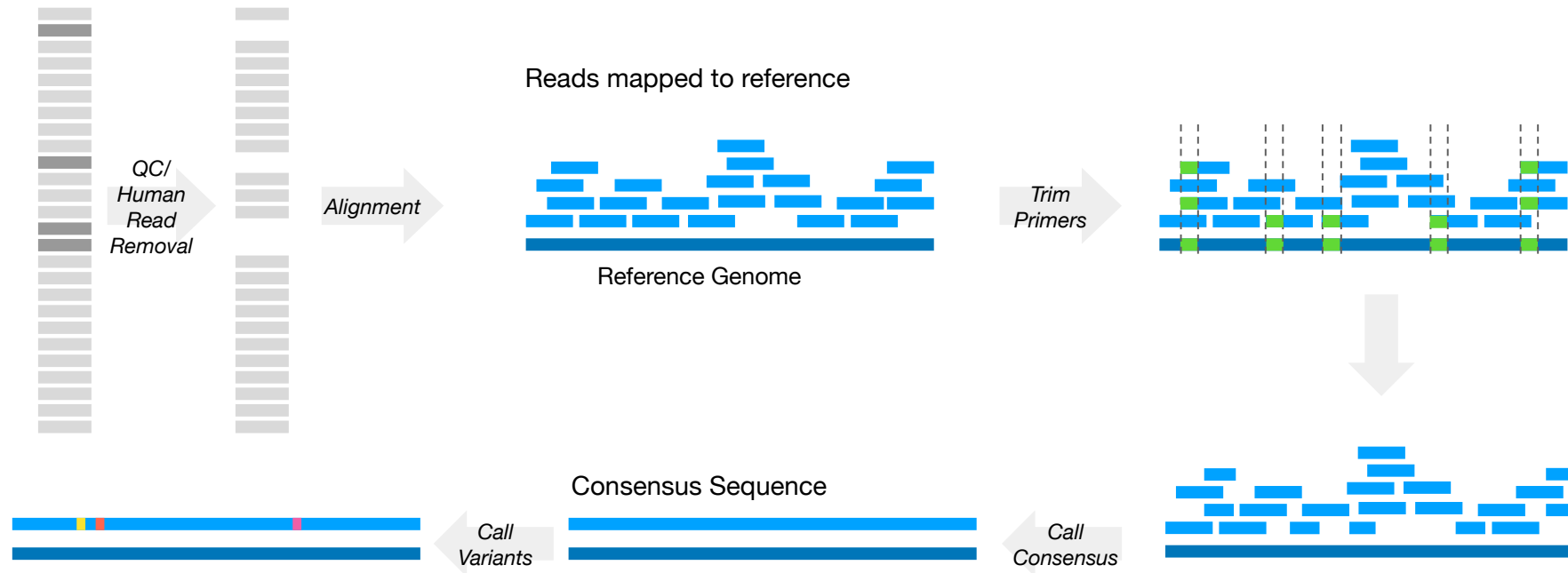


Consensus Genome Sequence

- Software for calling consensus: [iVar](#)
 - Minimum depth to call bases:
 - Amplicon data (BED file provided): 10 reads
 - mNGS data (no BED file): 5 reads
 - Quality threshold: Q20
 - Frequency threshold: 0.75 (to call a specific base, it should be observed at > 75% frequency)

Genome Assembly Pipeline

Raw reads



Notes:

- Variant calling software: [BCFtools and samtools](#)

Genome Report

CI >

sample1_2 ▾

[Sample Details](#)

[Download All](#)

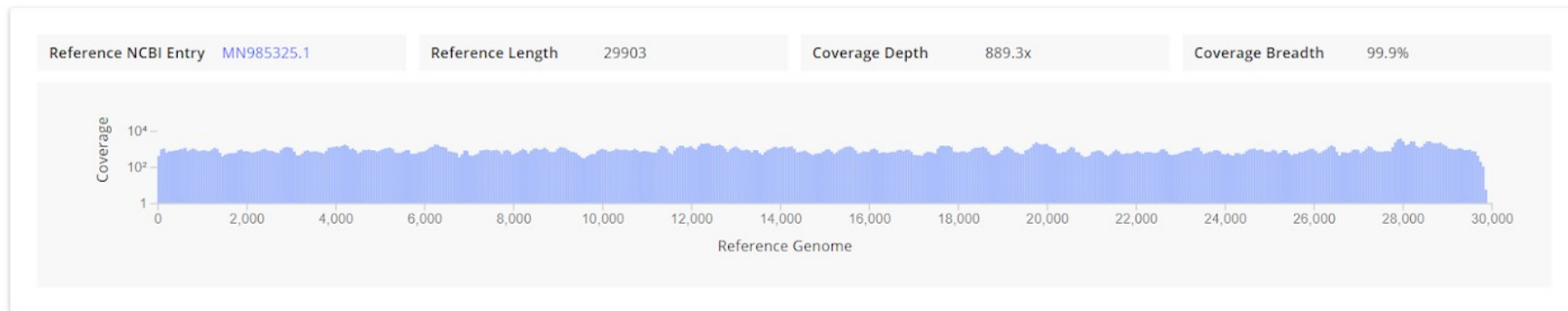
Consensus Genome BETA

[Learn more about consensus genomes >](#)

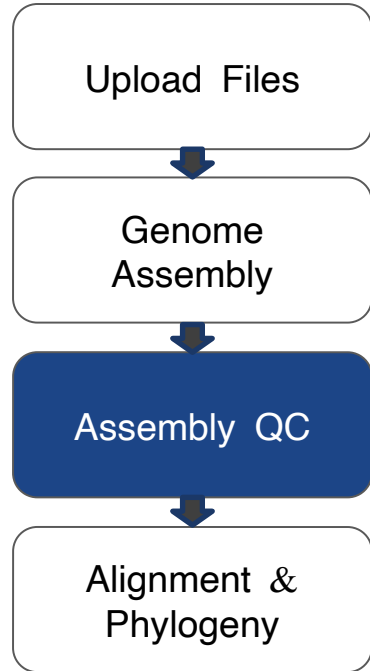
Is my consensus genome complete? ⓘ

Taxon	Reads	GC Content	SNPs	%id	Informative Nucleotides	Missing Bases	Ambiguous Bases
Severe acute respiratory syndrome coronavirus 2	187444	38.01%	7	100%	29850	12	0

How good is the coverage? ⓘ



Workflow for Generating Consensus Genomes



Can I use the assembled consensus genome for downstream analyses?

QC Checks

CI >

sample1_2 ▾

[Sample Details](#)

[Download All](#)

Consensus Genome BETA

[Learn more about consensus genomes >](#)

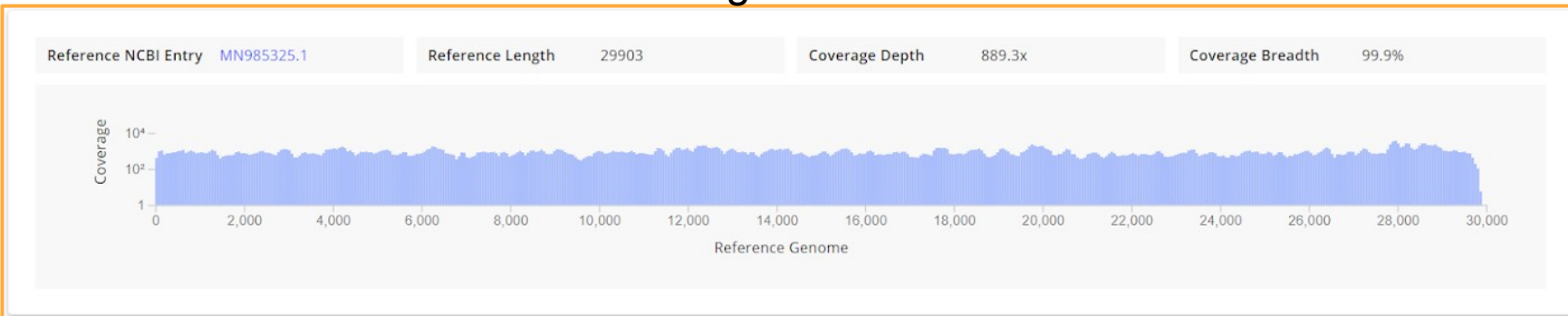
Is my consensus genome complete? ⓘ

Assembly Metrics

Taxon	Reads	GC Content	SNPs	%id	Informative Nucleotides	Missing Bases	Ambiguous Bases
Severe acute respiratory syndrome coronavirus 2	187444	38.01%	7	100%	29850	12	0

How good is the coverage? ⓘ

Coverage Stats



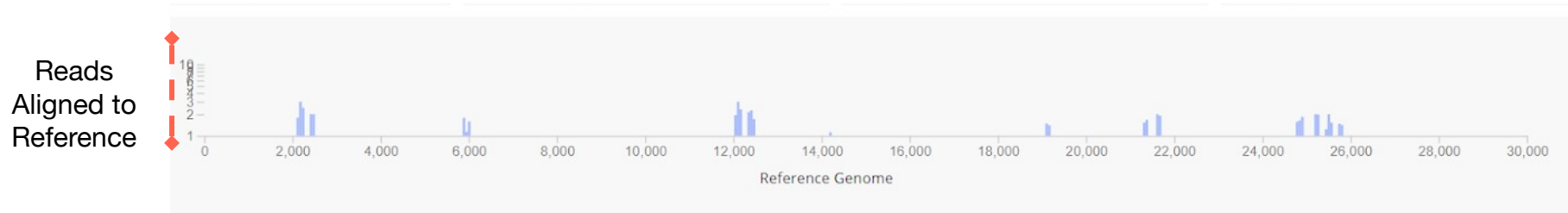
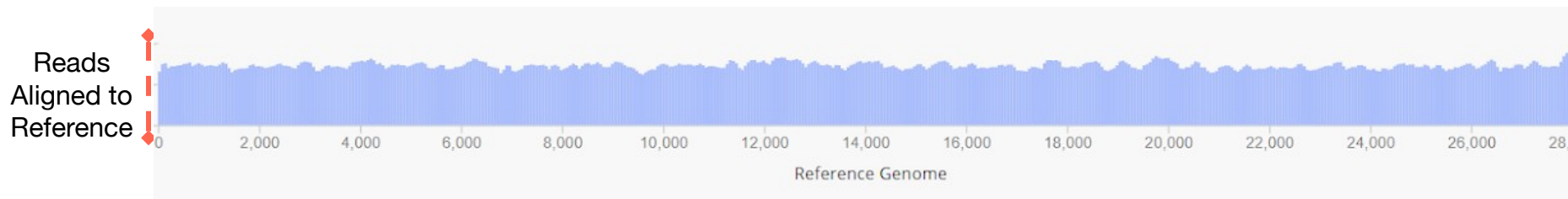
Genome QC: Assembly Metrics

- **Mapped Reads** - Number of reads aligning to the reference accession.
- **GC Content** - Percentage of bases in the consensus that are either guanine (G) or cytosine (C) (should be close %GC of the reference sequence).
- **SNPs** (single nucleotide polymorphisms) - Represent single nucleotide variations between the reference accession and consensus genome.
- **% Identity (id)** - Percentage of nucleotides in the consensus genome that are identical to those in the reference accession.

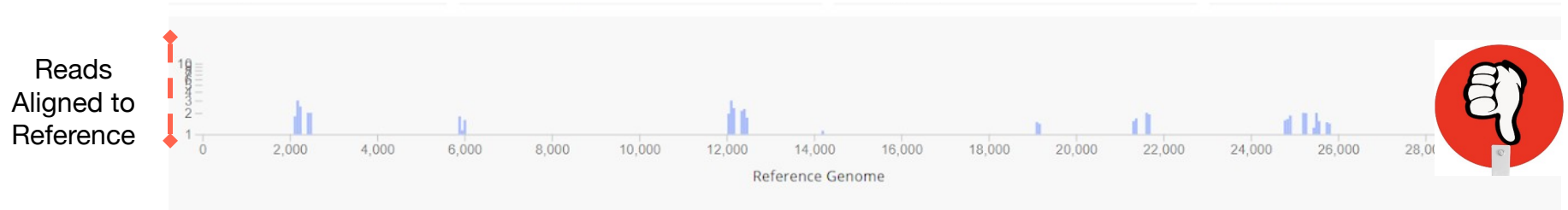
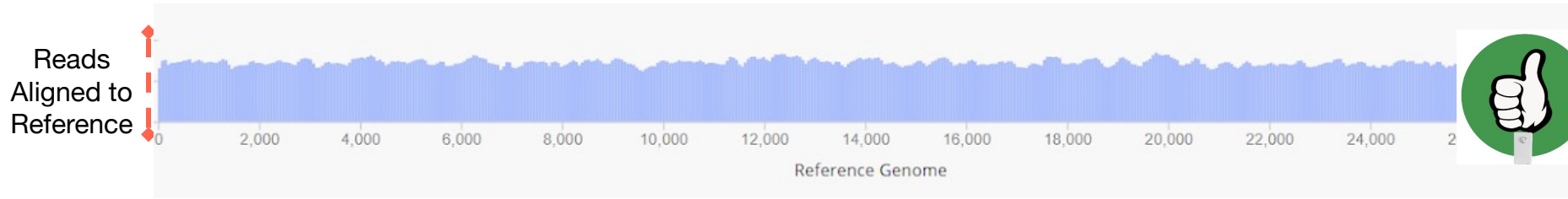
Genome QC: Assembly Metrics (Continued)

- **Informative Bases** - Number of nucleotides that are A,T,C, or G. CZ ID's consensus genome pipeline only calls nucleotides supported by at least 5 (mNGS data) or 10 (amplicon data) reads.
- **% Genome Called** - Percentage of the genome meeting thresholds for calling consensus bases. The closer this number is to 100%, the better.
- **Missing Bases** - Number of Ns in the consensus genome.
- **Ambiguous Bases** - Number of degenerate bases (non-C, T, G, or A) in the consensus genome. CZ ID's consensus genome pipeline only calls nucleotides that are detected at least at 75% frequency.

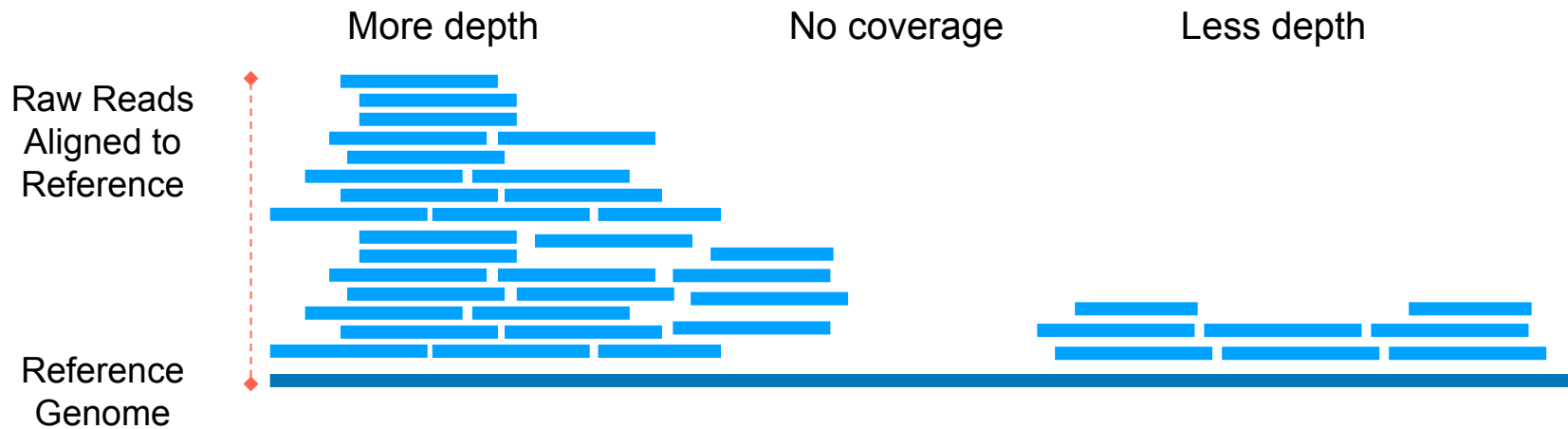
The coverage plot is a great first QC check



The coverage plot is a great first QC check



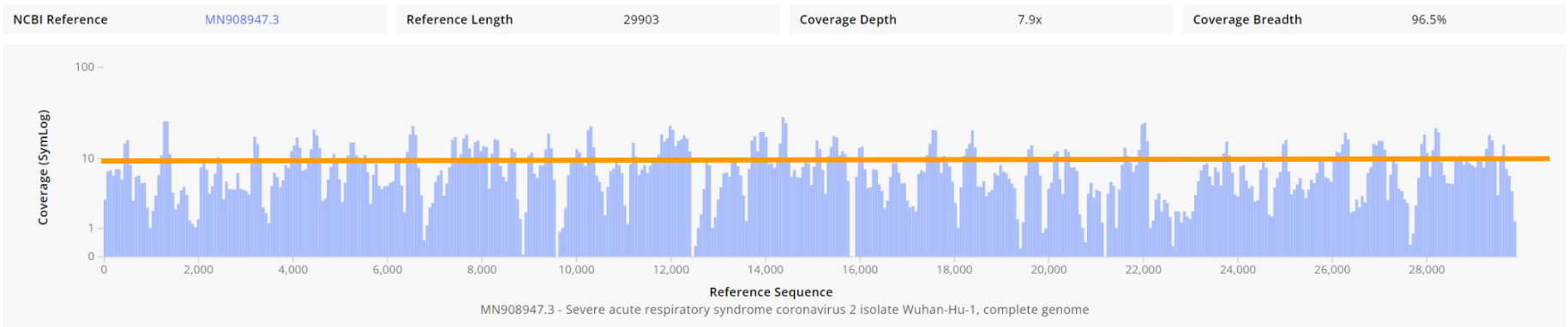
Is there enough coverage depth?



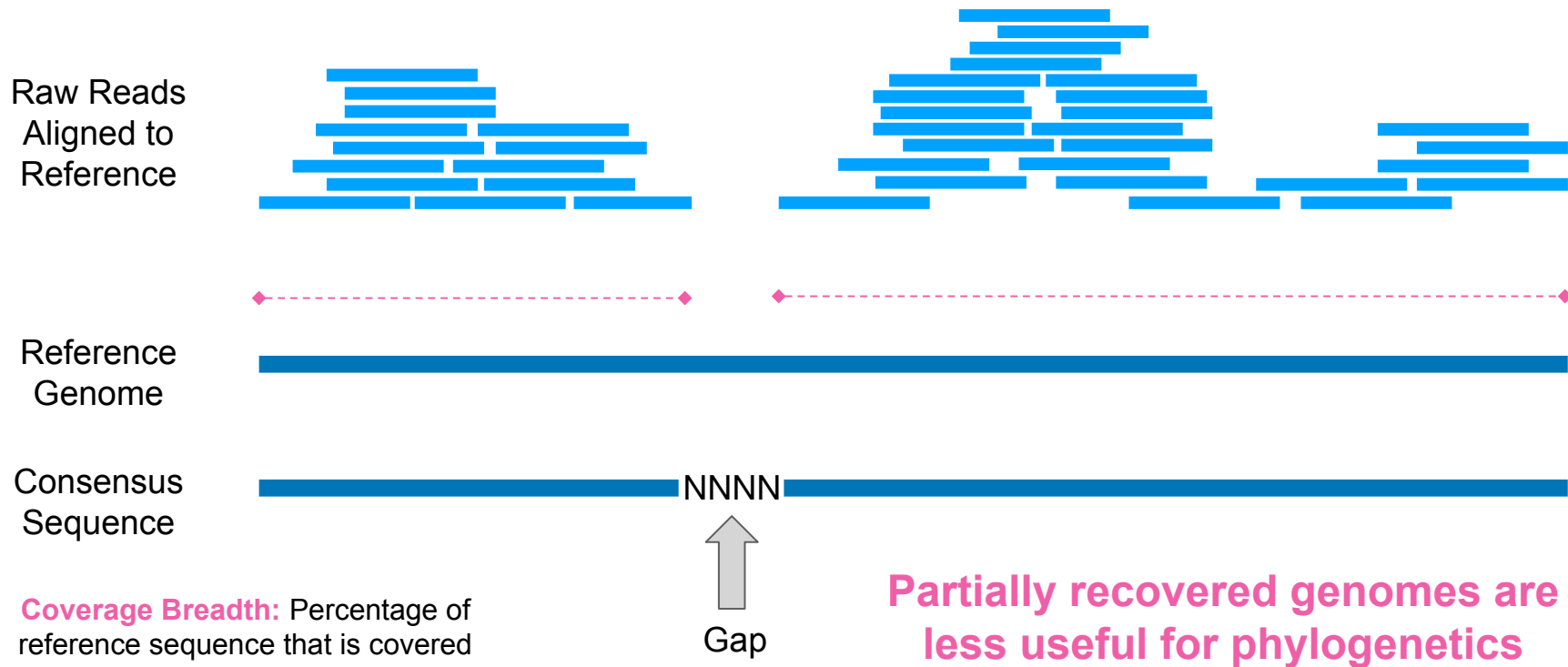
Coverage Depth: # of times a nucleotide is read during sequencing

Coverage Depth Matters

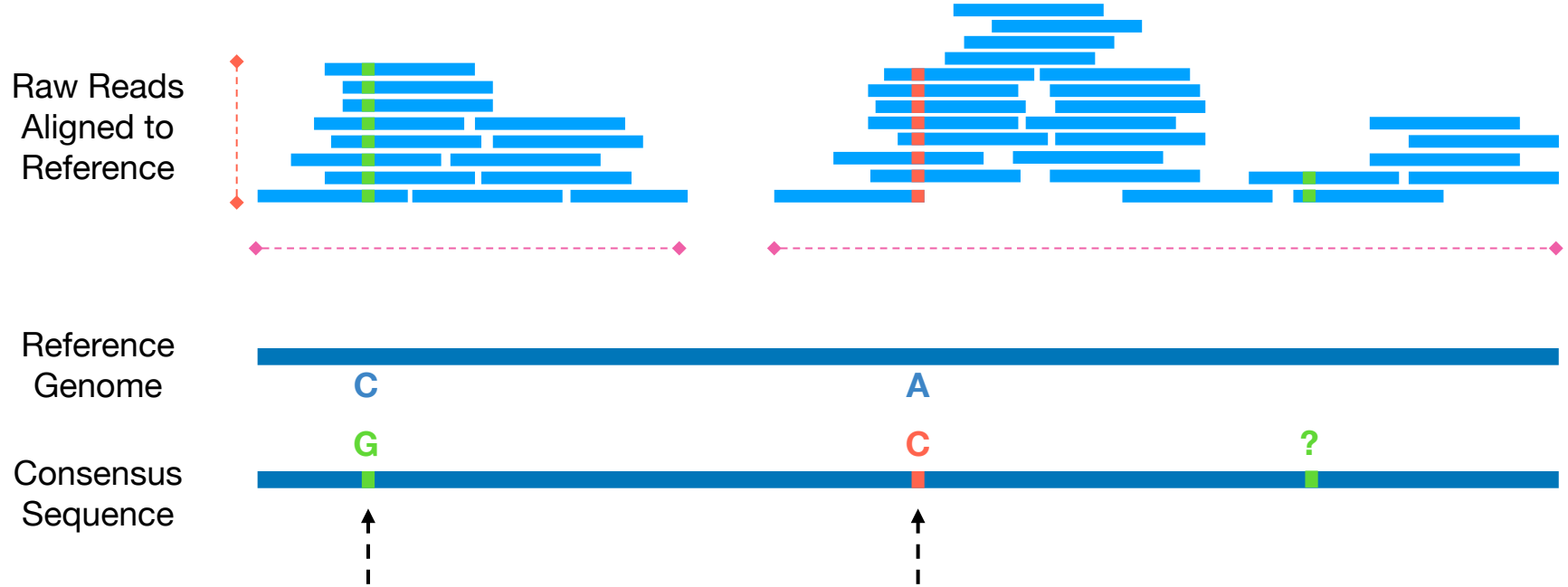
- For WGS data, any area with coverage of **< 10 reads will be called an N** in the consensus genome (< 5 reads will be called an N for the mNGS consensus pipeline)
- The consensus genome below would have poor quality (most base calls would be Ns)



Was most of the genome recovered (coverage breadth)?



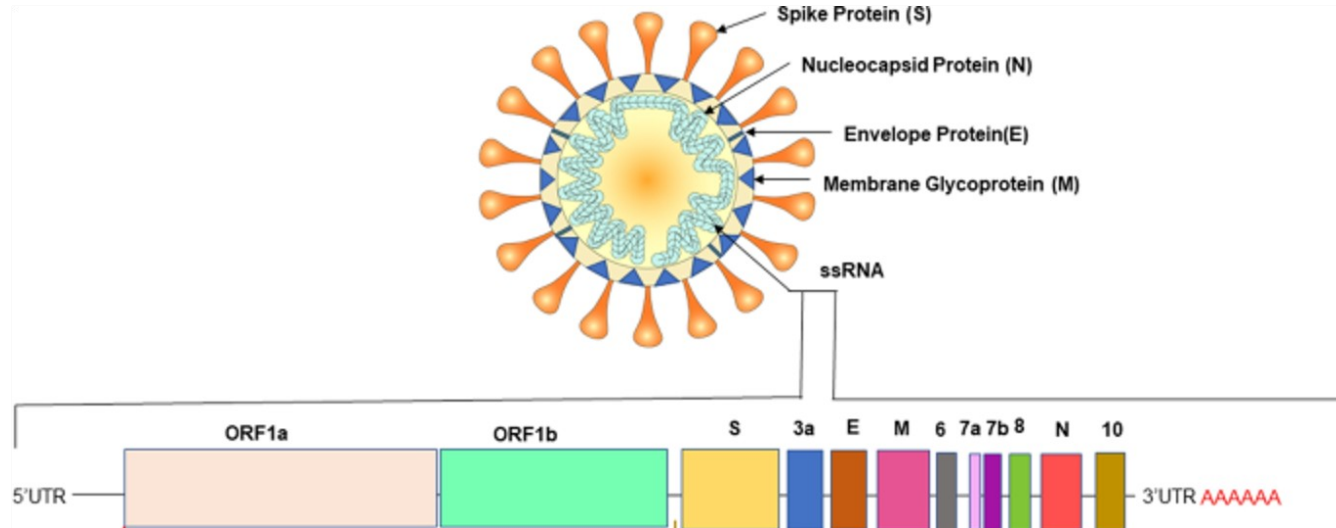
Evaluating Coverage to Identify SNPs



SNPs: Single Nucleotide Polymorphisms, variations in a single base pair in a DNA sequence. The number of acceptable SNPs is pathogen dependent

What's next?

- Does the genome include predicted coding regions?
 - Find open reading frames (ORFs) and compare to known genomes



What's next?

- Does the genome include predicted coding regions?
 - Find open reading frames (ORFs) and compare to known genomes
- Which serotype or variant does it represent?
 - Use available typing tools to determine the virus serotype or lineage

[Nextclade](#) - SARS-CoV-2, Influenza (Flu), Mpox (Monkeypox), Respiratory Syncytial Virus (RSV)

[Genome Detective](#) - Arboviruses, hepatitis viruses, enteroviruses, measles virus

- Phylogenetic analysis